# Sougandhi Manonmani Pendyala

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## PROFESSIONAL SUMMARY

Software Development Engineer with Amazon experience in building scalable full-stack applications, distributed systems, and cloud-native services using Java, Spring Boot, React, TypeScript, and AWS. Works across event-driven architectures, REST APIs, automated testing, and operational monitoring, grounded in data structures and algorithms.

## PROFESSIONAL EXPERIENCE

**Software Development Engineer**
**Amazon**
Detroit, MI, USA
Sep 2025 - Oct 2026

- Re-architected offer expansion from synchronous request-blocking to an event-driven SQS FIFO pipeline processing 1,000+ messages/day: built the ordered consumer and filter-chain routing, with DLQ recovery and tiered CloudWatch alarms provisioned via AWS CDK.
- Built policy-driven rate limiting and traffic shaping on AWS AppConfig, defined in code and CDK, giving on-call a config-push lever to change rate limits across regions in minutes, work that previously required a full service deployment.
- Shipped a self-service BIL migration workflow using Spring Boot REST APIs and React with merchant access validation, including migration-intent APIs and a localized in-app banner, replacing an ops-mediated process; achieved 100% unit coverage with Vitest.
- Wrote a connection-removal script with a dry-run mode that safely retired 2M unproductive merchant connections, preceded by a 30-day communication campaign reaching 5M sellers.
- Built the Expansion Preferences page end-to-end in React with Zustand and launched it behind an A/B experiment, enabling self-service configuration of multiple offer expansion preferences and reducing turnaround time by 25%.
- Built a React cooldown mechanism with localStorage persistence and polling hooks, preserving workflow state across refreshes/session breaks and reducing duplicate submissions by 40%.
- Carried on-call for the service, resolving incidents, driving RCAs, and adding diagnostic logging and alarms that reduced time-to-diagnose by 30% for recurring failures.
- Cut Java microservice cold-start time by 33% (90s to 60s) through improvements across 3 layers: JVM tuning, startup automation, and deployment pipelines.
- Used Claude and Kiro across 4 development workflows (implementation, test generation, debugging, and iteration), halving feature turnaround from 2 weeks to 1 while maintaining production-quality code.

## PROJECTS

**SafeSend:** Built a hybrid AI moderation system that blocks abusive messages before they are sent: an on-device text normalizer and multilingual word list (~40 languages) cascades into a zero-shot, multi-label Claude classifier (20 categories) with JSON-schema structured output, prompt-injection defence, explainable reasons, and fail-closed handling; added human-in-the-loop appeals and a 41-message dialect-fairness evaluation.
JavaScript, Node.js, HTML, Claude API

**Improving Software Quality Framework Using AI Platforms:** Published a peer-reviewed book chapter on AI-driven software QA: defect detection, risk prediction, workflow automation, Agile/DevOps.

**Mail Scheduler Agent:** Built a Claude agent that schedules emails from plain English; Supabase Edge Functions + pg_cron send via Gmail with retries.
Python, FastAPI, Supabase, Claude, Gmail API

**Platepix3D:** Built a prototype turning dish photos into 3D menu models via MobileNet and react-three-fiber; added inventory forecasting.
Next.js, TypeScript, TensorFlow.js, Three.js, Tailwind

**Multi Language AI Chatbot:** Engineered a 5-language chatbot using OpenAI and Google Translate APIs, with secure authentication, and persistent chat history for user interactions.
Python, Streamlit

**Pawcare:** Developed a pet care app for grooming, vet appointments, and pet-sitting with scheduling and reminders; integrated Google Maps, Firebase, and Calendar APIs.
Java, Android Studio

## EDUCATION

**Master of Information Technology and Management**
**Illinois Institute of Technology**
Aug 2023 - May 2025
CGPA: 4.0/4.0

**Bachelor of Technology in Electronics and Communication Engineering**
**Kamala Institute of Technology and Science**
CGPA: 3.0/4.0

## SKILLS

**Languages:** Java, TypeScript, JavaScript, Python, SQL, HTML, CSS

**Full Stack:** React, Redux, TypeScript, Spring Boot, REST APIs, Microservices, Distributed Systems

**Cloud & AWS:** AWS CDK, SQS, DynamoDB, CloudWatch, IAM, AppConfig

**AI & LLMs:** Claude API, OpenAI API, Tool Calling, AI Agents, Prompt Engineering, TensorFlow.js

**DevOps & Tools:** Docker, Kubernetes, CI/CD, Git, GitHub, GitLab, Jira, Claude, Kiro

**Testing:** JUnit, Mockito, Vitest, Integration Testing, Functional Testing

**Core CS:** Data Structures, Algorithms, Object-Oriented Programming